

Time as Emergent from Memory Topology

Leslie Yarbrough

December 2025

Abstract

Time is commonly treated as a fundamental dimension along which physical, biological, and cognitive processes evolve. We propose an alternative framework in which time is emergent, arising from the topological structure of memory within a system. Memory is modeled not as stored content or retrievable states, but as a constraint topology that shapes permissible state transitions. Temporal experience corresponds to the ordering relation induced by traversal of this topology. We show that dreaming provides a natural demonstration of temporality without external clocks or sensory input, and we introduce a laminar-flow model of memory propagation in which historical structure is embedded in state evolution rather than explicitly recalled. This framework yields testable implications for neuroscience and artificial systems while remaining agnostic to metaphysical interpretation.

1 Introduction

In most theoretical frameworks, time is assumed rather than derived. Physics treats time as a coordinate; neuroscience treats it as a sequencing parameter; cognitive science often treats it as subjective duration. Despite differences in emphasis, these approaches share a common assumption: *time is a primitive within which systems operate.*

This paper challenges that assumption.

We propose that time emerges from the topology of memory—specifically, from how a system constrains, associates, and propagates internal states. Under this view, time is not an external axis but a functional ordering induced by memory structure. Temporal continuity reflects coherence of state transitions rather than motion along a preexisting dimension.

Notation

- $\mathcal{S}$: State space of the system
- $s \in \mathcal{S}$: A complete internal configuration
- $\mathcal{M} \subseteq \mathcal{S} \times \mathcal{S}$: Memory topology defining admissible transitions
- $\prec$: Emergent temporal ordering relation induced by $\mathcal{M}$
- τ : Internal progression parameter indexing state evolution

2 Memory as Topology, Not Storage

Standard models conceptualize memory as a repository: discrete stored representations, indexed retrieval mechanisms, and explicit access to past states. We instead define memory as topological constraint.

2.1 Formal Definitions

Definition 1 (State Space). Let $\mathcal{S}$ denote the state space of a system, where each $s \in \mathcal{S}$ represents a complete internal configuration.

Definition 2 (Memory Topology). Let memory be represented by a constraint manifold $\mathcal{M} \subseteq \mathcal{S} \times \mathcal{S}$ defining admissible transitions between states. The topology of $\mathcal{M}$ determines which transitions are reachable, stable, or suppressed.

Under this definition, memory:

- need not be accessed explicitly,
- need not encode content symbolically,
- operates by shaping the geometry of possible future states.

Memory therefore functions structurally rather than archivally. Its role is not to represent the past, but to constrain the future.

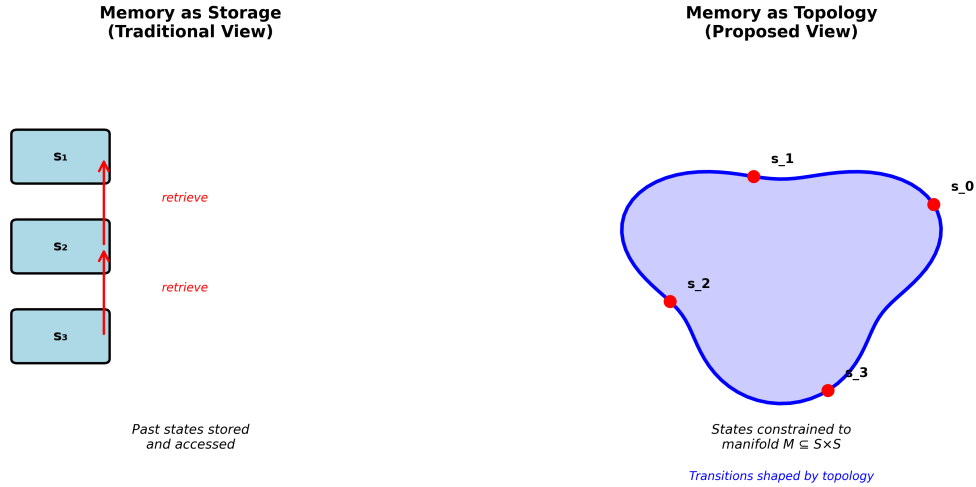


Figure 1: Comparison between classical storage-based memory (left) and topological memory (right). In the proposed model, memory constrains admissible transitions rather than storing retrievable content.

3 Time as Emergent Ordering

If memory defines permissible transitions, then time emerges as the ordering induced by those transitions.

Definition 3 (Emergent Time). Let $\prec$ be a partial ordering on $\mathcal{S}$ such that:

$$s_i \prec s_j \iff s_j \text{ is reachable from } s_i \text{ under } \mathcal{M} \quad (1)$$

Under this formulation:

- Time is not identified with a real-valued parameter $t \in \mathbb{R}$

- Temporal order corresponds to associative reachability
- Temporal continuity corresponds to coherence of successive transitions

What is experienced as “time passing” is the maintenance of ordered association across evolving states. There is no external clock—only the structure carried forward.

4 Dreaming as Internal Temporal Rendering

Dreaming provides a natural test case for emergent time. During dreaming:

- No external sensory input is present
- No physical photons are processed
- No objective clocks are referenced

Yet dreams exhibit ordered sequences, causal structure, duration, and visual experience. Let dream-state evolution be modeled as:

$$s_{n+1} = F(s_n, \mathcal{M}, E) \quad (2)$$

where F is an internal transition function, $\mathcal{M}$ is memory topology, and E represents affective or associative weighting.

Temporal structure arises without reference to an external time variable. This demonstrates that temporal experience can be fully rendered by internal ordering alone, supporting the claim that time is emergent rather than fundamental.

Neuroscientific evidence shows that during REM sleep, the visual cortex synthesizes light experience from memory, emotion, and association—constructing phenomenological reality without external photons. This synthesis operates through the same memory topology $\mathcal{M}$ that governs waking state transitions, but with different affective weights E and reduced sensory constraints.

5 Laminar Flow Model of Memory Propagation

We propose that memory propagation operates under a laminar flow principle, contrasting with turbulent retrieval-based models.

Definition 4 (Laminar Memory Flow). A system exhibits laminar memory flow if its internal state evolution satisfies:

$$\frac{\partial^2 s}{\partial \tau^2} \approx 0 \quad (3)$$

where τ is an internal progression parameter indexing successive state updates.

This condition indicates smooth propagation of structure without abrupt retrieval, overwrite, or discontinuous jumps. In laminar systems:

- History is embedded in the current state
- Memory is implicit rather than explicit
- Long-horizon coherence is preserved without global lookup

This contrasts sharply with turbulent memory models, in which past states must be explicitly accessed or recombined. Recent advances in AI architectures (state-space models, linear RNNs) demonstrate that laminar flow enables stable long-context processing without the quadratic costs of attention mechanisms—a computational validation of this theoretical principle.

6 Observer-Dependent Temporal Rendering

Because memory topology $\mathcal{M}$ differs across systems, temporal rendering is observer-dependent. Two systems exposed to identical external stimuli may experience different temporal orderings due to differences in:

- Constraint topology
- Associative weighting
- Affective modulation

This does not imply subjectivism about physical reality, but rather that temporal experience is relational, emerging from internal structure rather than imposed by a universal clock. The ordering $\prec$ is real and functional, but it is not absolute.

7 Testable Implications

This framework yields concrete, falsifiable predictions.

7.1 Neuroscience

Prediction 1. Disruption of associative topology—via targeted modulation of associative pathways rather than ablation of content-specific regions—should non-linearly distort perceived temporal ordering $\prec$, even when memory content remains intact.

Prediction 2. Subjective dream duration should correlate more strongly with associative density (the number of ordered transitions per unit τ) than with global neural firing rates.

7.2 Cognition

Prediction 3. Trauma-induced temporal distortion arises from deformation of the transition manifold $\mathcal{M}$, introducing unreachable or cyclic regions. Therapeutic interventions restore coherence by reshaping $\mathcal{M}$ to support laminar state propagation:

$$\frac{\partial^2 s}{\partial \tau^2} \approx 0 \tag{4}$$

Prediction 4. Narrative reframing restores temporal continuity not by altering memory content, but by restructuring the topology that constrains future state access.

7.3 Artificial Systems

Prediction 5. Systems designed with flow-based memory architectures (e.g., state-space models) should exhibit stable long-horizon temporal coherence without explicit clock variables. Temporal ordering $\prec$ should emerge even when no external time parameter $t \in \mathbb{R}$ is referenced.

8 Philosophical Implications

This framework dissolves apparent contradictions between claims that “information is fundamental” and “information is illusory.” Information structures are emergent artifacts of constrained flow through $\mathcal{M}$, not ontological primitives. Time is not denied—it is derived.

The box analogy clarifies this: memory is not a container to be opened and searched, but a physical object whose *shape* determines what actions are possible. We use memory by climbing on it, pushing past it, escaping through windows it makes accessible. The topology matters; the contents do not require explicit access.

9 Scope and Limitations

This framework focuses on the functional emergence of subjective temporal experience in systems defined by a memory topology $\mathcal{M}$, yielding a derived, observer-dependent temporal ordering $\prec$. However, this framework does not make claims regarding:

- **Physical Time:** This framework is agnostic to the nature of fundamental physical time t as treated in general relativity or quantum mechanics. The topological ordering $\prec$ operates within a system’s internal state space $\mathcal{S}$ and explains subjective temporal rendering, not objective spacetime structure.
- **The Hard Problem of Consciousness:** While temporal ordering is a core component of phenomenal experience, this model addresses the functional mechanism of ordering, not the qualia or subjective feeling of “time passing.”
- **Causality:** The ordering $s_i \prec s_j$ establishes *associative reachability*, which serves as the cognitive basis for perceived causality. This framework does not provide a new theory for physical causality at the ontological level.

The central claim is restricted to the derivation of temporal experience from structural memory constraints, without requiring the external time coordinate t as a primitive for cognition.

10 Conclusion

Time is not a container within which memory operates. Memory is the structure from which time emerges.

By modeling memory as topology and time as ordering, we obtain a coherent account of temporal experience grounded in internal dynamics rather than external dimensions. This reframing offers a unifying perspective across cognition, neuroscience, and artificial systems without requiring time as a fundamental ontological primitive.

The dream does not unfold in time. Time unfolds from the dream’s internal structure.